Limitations of reptation theory for modeling the stress-dependent rheological behavior of Polyethylene Terephthalate above the glass transition temperature

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Limitations of reptation theory for modeling the stress-dependent rheological behaviour of Polyethylene Terephthalate above the glass transition temperature

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Abstract

The biaxial stretch blow moulding is an established process for manufacturing plastic containers, in which preforms are stretched both in circumferential and axial directions while being blown into a mould. In the development phase of these products, computer-aided analysis tools are extensively used to increase the material and process efficiency. The accuracy of these tools depends on the underlying material models and parameters.

The aim of this paper is to investigate the suitability of reptation theory for the prediction of the strain-dependent rheological behaviour of polyethylene terephthalate (PET) in the stretch blow moulding process. Reptation theory has already been successfully applied to a number of polymer melts in the past decades. However, the practical applicability of reptation theory for predicting the strain-dependent rheological behaviour of highly viscous polymers slightly above the glass transition temperature, as is the case with stretch blow moulding, has not yet been fully investigated.

In the first step, the constitutive material model equation of reptation theory is implemented and the necessary model parameters are determined using various measurement methods. However, the measurements could not be conducted with the same accuracy as in the case of polymer melts, since the measurement methods used showed instabilities in the glass transition temperature range, which led to high measurement uncertainties. Consequently, the application of the material model does not match quantitatively to biaxial stretch tests. Qualitatively, on the other hand, the material model successfully reproduces the stress-strain behaviour of PET films at low strains. In case of temperature dependence, the model results are neither qualitatively nor quantitatively satisfactory.

The temperature dependency of the material model has been further investigated in the second step. It was shown that the derivative of the Doi-Edwards memory function with respect to the temperature has an inflexion point if the stretching duration is equal to the disengagement time. For very small disengagement times compared to the stretching duration, the results of the model match the experimental observations. For high disengagement times induced by the large viscosities near the glass transition temperature and for low stretching times induced by high strain rates, however, the Doi-Edwards memory function cannot predict the experimental observations correctly.

The investigations show that reptation model qualitatively predicts the strain behaviour of biaxial stretched PET films at low strains correctly. However, different measurement approaches for a more accurate and reproducible determination of the material properties and a modification of the model are required in order to adapt the model to highly viscous melts above the glass transition temperature. The results have shown that the process conditions of the two-stage stretch blow moulding, such as high strain rates and low processing temperatures, exceed the validity limits of reptation theory.

Introduction

Plastic containers made from PET are widely used in the packaging of beverages and cosmetics. The most common manufacturing process for the large-scale production of these containers is the stretch blow moulding process as shown in Figure 1, in which plastic containers are manufactured by stretching a preform in circumferential and axial directions while being blown into a mould [1]. Before the the forming station, the preforms are heated using infrared radiation emitted by tungsten filaments to a temperature between 90 and 110 °C, which is slightly above the glass transition temperature of the material. Figure 2 shows exemplary the temperature profile of a PET preform prior to the forming station.

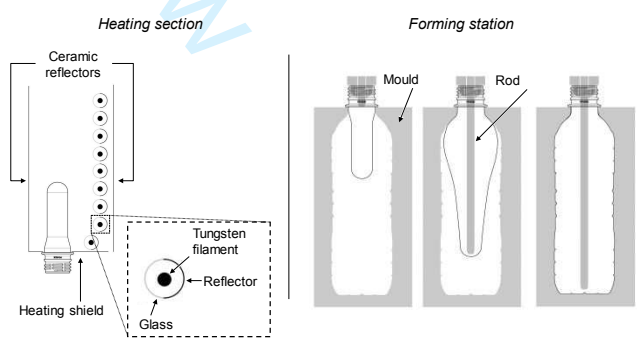

Fig. 1: Two-stage stretch blown mould

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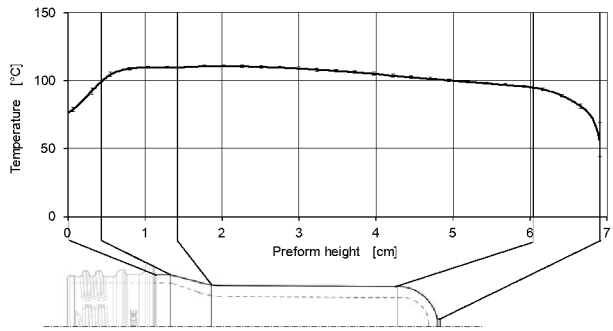

Fig. 2: Temperature profile of a PET preform prior to the forming station

Computer-aided analysis (CAD) tools are extensively used to improve the mechanical properties of these products in the design phase as well as to increase the cost efficiency of the manufacturing process. The accuracy of the CAD tools depends on the underlying material models and parameters. In the past decades, a number of material models have been developed for the prediction of the strain-dependent rheological behaviour of amorphous and semi-crystalline polymers.

The elongation of the polymer during the stretch blow moulding process leads to a strong orientation of the molecular chains and to the formation of crystalline lamellar structures, which in turn increase the material strength [2, 3]. The polymer is cooled rapidly after the stretching process and thus, the molecular orientations are retained [4]. The stress resulting from biaxial strain depends at least on the stretch ratio, the molecular weight, the material temperature and the strain rate [5].

The stress-strain behaviour of PET during elongation is illustrated qualitatively in Figure 3. At low strains, the stress-strain curve follows Hooke's law. After a plateau, the stress increases slightly with increasing strain. At high strains, the strain hardening plays an important role, leading to a quasi-exponential increase of the stress. Summarised, the stress caused by biaxial strain is highly non-linear.

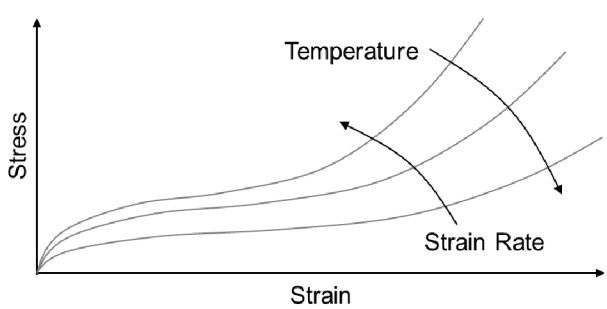

Fig. 3: Stress of a PET film caused by biaxial strain for different strain rates and temperatures [4]

Depending on the requirements, a number of material models have been used in simulations of the stretch blow moulding process. Hyperelastic models (e. g. Mooney-Rivlin, Neo-Hooke, Ogden or Yeoh models) predict the rubber-like behaviour of PET and strain hardening at high strains [6]. An extension through a time-temperature shift and a Prony series considers the dependency of the stress-strain behaviour on the temperature and the strain rate [7]. Viscoplastic and elastoviscoplastic material models (e. g. creep models) predict the time-dependent irreversible plastic deformations [8, 9]. However, viscoplastic models neglect the elastic material deformations and the strain hardening. Elastoviscoplastic models contain a linear elastic component which allows the prediction of the material behaviour over a larger temperature range [10, 11]. Viscoelastic material models are generally based on a Maxwell spring-damper model. Due to this approach, the dependency of the strain rate on the temperature is considered, however, not the strain hardening at high strains [12, 13]. Viscohyperelastic material models combine the advantages of the hyperelastic models in the case of a dependence on temperature and strain rate and viscoelastic models in the case of an accurate description of the strain hardening. The glass-rubber model according to Buckley et al. with an extension according to Adams et al. is a viscohyperelastic model for the prediction of stress behaviour of PET films caused by the biaxial elongation [14, 15, 16].

Hyperelastic, viscoelastic and viscohyperelastic approaches achieve promising results for the description of the material behaviour in the stretch blow moulding process. The coefficients of these models can be obtained from fits based on biaxial stretch tests. Although these models provide a good prediction of the material behaviour in the temperature and strain rate range regarded in the experimental tests, they cannot guarantee the accuracy of predictions outside of this range.

Reptation theory according to De Gennes, which is based on the diffusional mobility of polymer chains, is an alternative approach for the prediction of the stress-strain behaviour of highly entangled polymer melts [17, 18]. Reptation model is a physically derived model and therefore should not be limited in its applicability to a calibration range, as long as the model assumptions can be applied. The model parameters are physically measurable quantities such as the storage modulus and the zero shear viscosity. However, reptation theory has so far only been used for polymer melts. An application for polymers slightly above the glass transition temperature has not yet been fully investigated.

The aim of this contribution is therefore to investigate the applicability of reptation model for predicting the stress-strain behaviour of biaxial stretched PET films slightly above the glass transition temperature at high strain rates, as in the case of stretch blow moulding process. At these temperatures, the storage modulus and the zero shear viscosity are much higher compared to melts. If the

predictive capabilities of reptation model at these boundary conditions are similar to those of melts, the model can predict the material behaviour more accurately in a wider temperature and strain rate range than the models used until now, without the need for a model calibration.

Reptation theory and governing equations

The concept of the polymer reptation according to De Gennes was first published in 1971 and describes the diffusion of polymer chains in a gel [17]. Doi and Edwards extended the theory for the modelling of entangled polymers and further investigations demonstrated the applicability of the theory for polymer melts [18-24].

The conformation of a polymer chain in an entangled state is described by Edward's tube model [25]. The polymer chain is confined in a tube-like region and the enveloping tubes of other macromolecules are considered as fixed obstacles, just like chemical and physical cross-links. Reptation theory describes the one-dimensional diffusional movement of a polymer chain within a tube as shown in Figure 4.

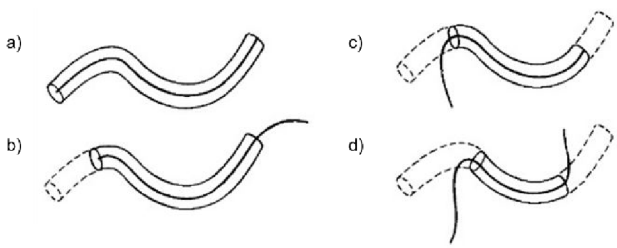

Fig. 4: Reptation of a polymer chain confined in a tube-like region [18]

The polymer chain is initially confined in a tubular area in which the fixed obstacles due to the surrounding macromolecules allow only a one-dimensional movement by diffusion (a). The parts of the polymer chain near the ends can leave the tube over time (b, c) and thus, the empty parts of the tube disappear (d). The time elapsed between the initial state and the completely vanished tube is specified as reptation or disengagement time τ_d, which is material-specific and temperature dependent [30].

$$\tau_d = \frac{20\ M_e}{\pi^2 R T \rho}\eta_0(T) \tag{1}$$

M_e: Molecular weight of entanglement
η_0: Zero shear viscosity
R: Universal gas constant
T: Temperature
ρ: Density

The fraction of the original tube that remains after a specific time t, which is also equal to the probability that the polymer chain is still confined in the original tube after the time t, is given by the Doi-Edwards memory function $\psi(t)$ [18].

$$\psi(t) = \sum_{p:\ \mathrm{odd}} \frac{8}{\pi^2 p^2} e^{-p^2 t/\tau_d} \tag{2}$$

t: Time
τ_d: Disengagement time

Reptation theory links the chain mobility due to one-dimensional diffusion with accessible material properties such as zero shear viscosity and molecular weight and makes an important contribution to the understanding of the rheological properties of highly entangled polymer melts [18].

In polymer melts, the stress is primarily a result of intramolecular forces and depends directly on the orientation of the bond vectors. As derived in [18], the stress tensor $\sigma_{\alpha\beta}(t)$ for dilute polymeric solutions is given through:

$$\sigma_{\alpha\beta}(t) = (\kappa_{\alpha\beta}(t) - \kappa_{\beta\alpha}(t))\eta_s + \sigma_{\alpha\beta}^{(P)}(t) + P\delta_{\alpha\beta} \tag{3}$$

κ: Velocity gradient tensor
η_s: Viscosity of the solvent
$\sigma_{\alpha\beta}^{(P)}$: Stress related to the chain confirmation
P: Pressure
$\delta_{\alpha\beta}$: Kroneker delta

The stress in diluted polymeric solutions is mainly due to the viscosity of the solvent, which is represented by the first term in Equation 3. As the polymer concentration increases, the polymer chains can be more easily oriented by the entanglements. This increases the contribution of the second term to the stress. At higher polymer concentrations, the stress related to the polymer chain confirmation dominates the first term. The stress contribution due to the orientation of polymer chains is given in Equation 4.

$$\sigma_{\alpha\beta}^{(P)}(t) = G_e \int_{-\infty}^{t} dt' \left(\frac{\partial}{\partial t'}\psi(t-t')\right) Q_{\alpha\beta}^{(IA)}(E(t,t')) \tag{4}$$

G_e: Plateau modulus
ψ: Doi-Edwards memory function
$Q_{\alpha\beta}^{(IA)}$: Orientation tensor (Universal tensor)
E: Deformation tensor

When applying Equation 3 to the biaxial strain of the polymer during the stretch blow moulding process, the viscous flow and the velocity gradients can be neglected, since the material is near the glass transition temperature and there is no solvent present in the material. Furthermore, the pressure inside the cavity during the blowing process does not apply in the strain direction and is subsequently

neglected. Hence, the remaining quantity is the stress resulting from the polymer chain itself.

The orientation tensor is related to the deformation of the polymer in the macroscopic scales and can be determined through a geometrical approach using the independent alignment (IA) approximation, without using any material parameters [22]. It denotes the orientation of the bond vectors within a chain [18].

$$Q_{\alpha\beta}^{(IA)} = \left\langle \frac{(E \cdot u)_\alpha (E \cdot u)_\beta}{|E \cdot u|^2} - \frac{1}{3}\delta_{\alpha\beta} \right\rangle_0 \quad (5)$$

E: Deformation tensor
u: Unit vector

The tensor $E(t,t')$ represents the deformation gradient at the time t related to the reference time t'. In case of an equibiaxial strain, both main directions are equally elongated and the deformation tensor takes the form assuming the incompressibility of the material:

$$E(t,t') = \begin{bmatrix} \lambda(t,t') & 0 & 0 \\ 0 & \lambda(t,t') & 0 \\ 0 & 0 & \lambda(t,t')^{-2} \end{bmatrix} \quad (6)$$

with

$$\lambda(t,t') = exp\left(\int_{t'}^{t} \dot{\varepsilon}(t'')\, dt'' \right) \quad (7)$$

λ: Stretch ratio
$\dot{\varepsilon}$: Strain rate

Equations 3 to 7 depend on the macroscopic deformation of the polymer and the material-specific parameters τ_d and G_e. The plateau modulus G_e is the storage modulus G' at a specific frequency, where the phase angle δ of storage and loss modulus is minimised corresponding to Equation 8.

$$G_e = G'(\omega)_{\tan \delta_{min}} \quad (8)$$

Equations 1 and 7 relate the model parameters to measurable and accessible quantities. The material model describes the rheology of various polymer melts satisfactorily, but has not yet been applied for polymers that are slightly above the glass transition temperature and are stretched at high strain rates.

Determination of the model parameters

As shown in the equations, the model requires the determination of density, molecular weight, plateau modulus and zero shear viscosity.

The PET type used is RAMAPET R180 of the company Indorama Ventures Public Company Limited, Bangkok, Thailand. The PET type is mixed with heating additives to increase the absorption of infrared radiation. The melting temperature is about 245 °C, the glass transition temperature is about 78 °C and the temperature range of the maximum crystal growth is about 140 °C.

The density of the PET was measured using a pvT100 measuring instrument of the company SWO Polymertechnik GmbH, Krefeld, Germany. The measured density is shown in Figure 5.

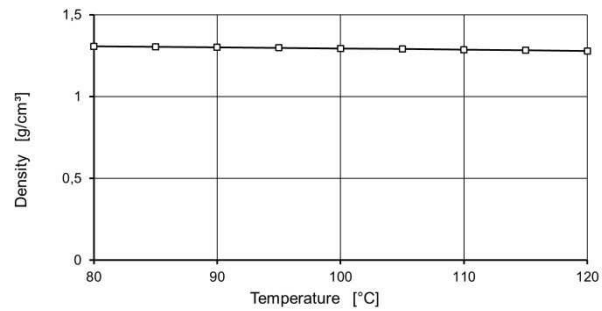

Fig. 5: Density of RAMAPET R180 near the glass transition temperature

Although the density profile decreases slightly with increasing temperatures, the density around the glass transition temperature is almost constant at 1.3 g/cm³. The practically constant density indicates that there is no phase transition in the investigated temperature range and thus, the polymer is in a thermodynamic equilibrium.

The storage and loss modulus were measured at 90 °C, 100 °C and 110 °C respectively, using the dynamic mechanical analysis (DMA) method with the measuring platform DMA Q300 of the company TA Instruments, Inc., New Castle, Delaware, USA. Specimens were taken from the body of the blow moulded PET bottles. The deviation between the measurements amounted to 40 %. Figure 6 shows four different measurements of the storage modulus at 100 °C.

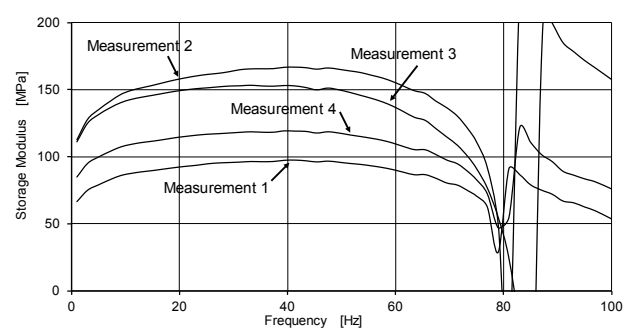

Fig 6: Measurements of the storage modulus of RAMAPET R180 at 100 °C

In addition to the high deviations, the measurements tend to be unstable at frequencies above 40 Hz. Since the

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DMA method is primarily designed for the measurement of solid elastic samples, the deviations and instabilities can be attributed to the partially viscous behaviour of the material above the glass transition temperature. The decreasing storage moduli at frequencies above 40 Hz indicate an increase of the material temperature due to frictional heating induced by the oscillating clamps. Comparing these results to the measurements of the storage moduli of melts in [27], the storage moduli show a plateau at a specific frequency slightly above 50 Hz, which is the minimum of the phase angle δ between the storage and loss modulus. This indicates that the plateau modulus corresponds approximately to the measured values at 40 Hz. Despite the high deviations, measurements were still carried out to determine the plateau modulus for different temperatures (Table 1). A temperature dependence of the plateau modulus cannot be determined from these data, therefore the plateau modulus is used as a constant parameter with 140 MPa in the following investigations.

Temperature [°C]	**Plateau modulus** [MPa]	**Deviation** [MPa]
90	142	36
100	134	27
110	142	47

Table 1: Average values of plateau modulus of RAMAPET R180 at different temperatures

The last remaining quantity necessary for the model equation is the zero shear viscosity. As illustrated in Figure 7, in polymers the viscosity converges to a limit at low shear rates, which is the zero shear viscosity. Measurements with the torsion rheometer under similar conditions as the dynamic mechanical analysis showed a linear increase of the measured viscosity for decreasing frequencies and no convergence to the zero shear viscosity.

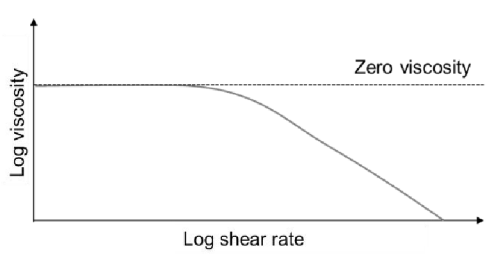

Fig. 7: Shear rate dependent viscosity in polymers

In order to investigate the model, the zero shear viscosity is obtained from a fit using the glass-rubber model in [14], where the zero shear viscosity is used for the calibration of a Maxwell spring-damper model. The zero shear viscosity η_0 is calculated through Equation 10.

$$\eta_0 = \eta_0^* \cdot exp\left(\frac{C}{T - T_\infty} - \frac{C}{T^* - T_\infty} + \frac{\Delta H_0}{RT} - \frac{\Delta H_0}{RT^*}\right) \quad (10)$$

C: Constant
μ_0^*: Viscosity in equilibrium at 67 °C
T_∞: Limiting temperature
T^*: Reference temperature
ΔH_0: Activation enthalpy

The values used were obtained in [14] from stress-strain curves of PET. As a result, the following values are used: $C = 2783$ K, $\mu_0^* = 5852\ GPa \cdot s$, $T_\infty = 276.8\ °C$ and $\Delta H_0 = 123\ kJ\ mol^{-1}$. The resulting zero viscosities are shown in Figure 8.

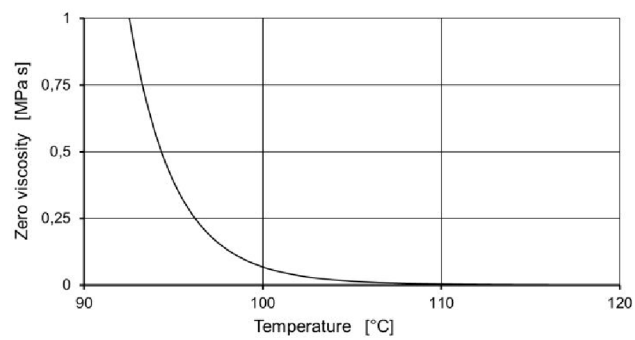

Fig. 8: Zero shear viscosity obtained from stress-strain curves of PET in [14]

The entanglement molecular weight M_e of PET is determined from [26] and is found to be 14,000 mol / kg.

In summary, the density can be obtained reproducibly by measurements near the glass transition temperature. The measurement of the plateau modulus shows instabilities in this temperature range, which leads to high deviations. For the zero shear viscosity, measurements with the torsion rheometer do not provide satisfying results. For this reason, the zero shear viscosity values from [14] are used to evaluate the model.

Evaluation of the model quality

The following investigations focus on the comparison of the stress-strain curves of PET films obtained from the model with measurements of biaxial stretching tests. Furthermore, the temperature and strain rate dependency of the constitutive equation are investigated. The biaxial stretching tests were performed at Blow Moulding Technologies, in Belfast, Northern Ireland. Reptation model was implemented with a plateau modulus of 140 MPa and disengagement times according to Equation 7 listed in Table 2.

Figure 9 and Figure 10 show the comparison of the model with the measured biaxial stress-strain curves at a strain rate of 1/s and temperatures of 100 °C and 105 °C.

Temperature [°C]	Zero shear viscosity [Pa·s]	Disengagement time [s]
100	68005.80	49.58
105	14188.21	10.21

Table 2: Zero viscosities and the corresponding disengagement times for the temperatures tested

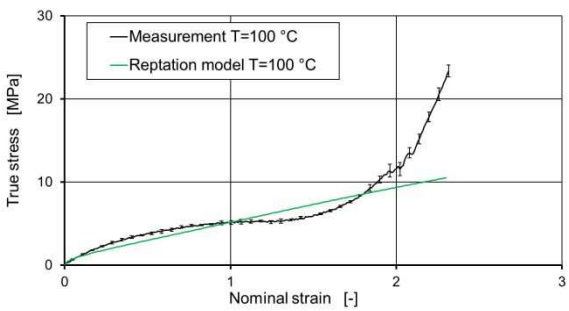

Fig. 9: Comparison between reptation model and measured biaxial stress-strain curves at 100 °C at a strain rate of 1/s

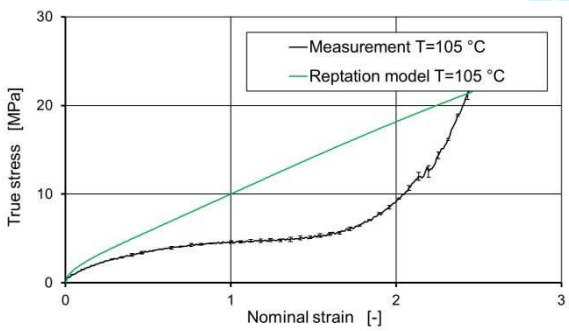

Fig. 10: Comparison between reptation model and measured biaxial stress-strain curves at 105 °C at a strain rate of 1/s

Figure 9 and Figure 10 show large differences in the models accuracy at different temperatures. The results at 100 °C indicate that reptation theory qualitatively predicts the biaxial stress-strain behaviour of PET slightly above the glass transition temperature at low strains correctly, up to an elongation of 100 %. However, the transition to the plateau and the strain hardening at high strains are not predicted by the model. The strain hardening is caused by interaction of a chain network [29]. Since reptation theory does not predict the strain hardening due to the molecular interactions, the further analysis focuses on low strains up to 100 %, where the strain hardening is not fully present.

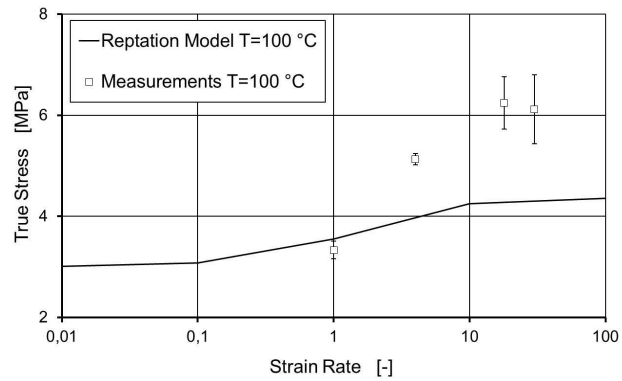

Fig. 11: Comparison of the strain rate dependency between reptation model and measured biaxial strain-stress at 100 °C at a constant strain of 100 %

Figure 11 shows the influence of the strain rate on the stress at a fixed strain of 100 %. The comparison of the strain rate dependency between the model and the experiments shows an agreement on the qualitative, but not on a quantitative level. The model basically reproduces the viscoelastic behaviour of polymers. The stress converges to a limit at high strain rates above 100 1/s, as well as at low strain rates below 0.01 1/s. However, the difference between these limits is significantly higher in the measurements compared to the model. In case of the comparison between reptation model and measured biaxial stress-strain curves at 100 °C in Figure 9, the results show an increasing deviation at different strain rates. The strain rate dependency is determined by the orientation tensor. The Doi-Edwards memory function $\psi(t)$ does not depend on the strain in this case, since the material data is not measured dependent on the strain rate.

Figure 12 shows the influence of the temperature on the stress at a fixed strain of 100%. An extract of the disengagement times is given in Table 3.

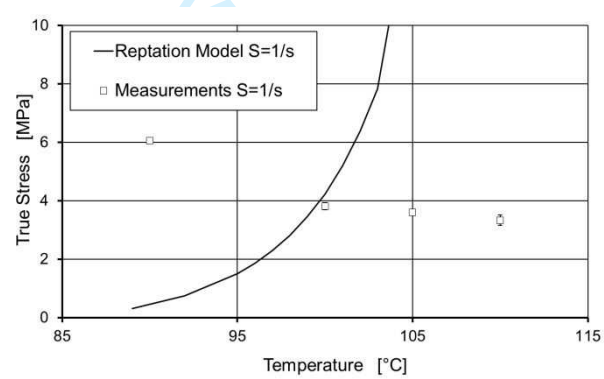

Fig. 12: Comparison of the temperature dependency between reptation model and measured biaxial strain-stress at a strain rate of 1/s at a constant strain of 100%

Temperature [°C]	Zero shear viscosity [Pa·s]	Disengagement time [s]
90	2820009.54	2112.99
100	68005.80	49.58
110	3434.88	2.44

Table 3: Disengagement times and zero viscosities of PET at different temperatures

It can be stated that the biaxial strain-stress behaviour of PET slightly above the glass transition temperature at high strain rates is neither qualitatively nor quantitatively predicted by reptation model in a satisfactorily way, since the exponentially increasing progression of the model is contrary to the experiments, where the true stress decreases with increasing temperatures.

Figure 13 shows the temperature dependency of reptation model at the same conditions as in Figure 12 but in a wider temperature range. The results show the model predicts a true stress maximum at 116 °C. At temperatures below 116 °C, the stress increases with increasing temperature, which does not correspond qualitatively to the measurements. Above the maximum, however, the increasing temperatures lead to a decrease of the true stress.

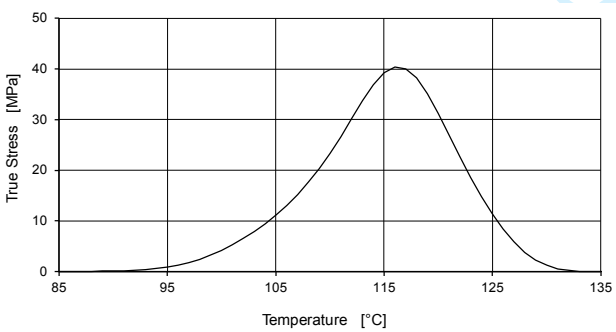

Fig. 13: Temperature dependency of reptation model at a strain rate of 1/s and a constant strain of 100%

Since the temperature dependence shown is not observed in the experiments, an extensive investigation of the temperature dependency of the stress is necessary.

Investigation of the temperature dependency of reptation model

The temperature dependence of the stress resulting from biaxial stretching is investigated by considering the derivative of the stress in Equation 4 with respect to the temperature. The experiments show that an increasing temperature leads to a decreasing stress value. The derivative of the stress with respect to the temperature must therefore be a negative value (Equation 11).

$$\frac{d\sigma}{dT} < 0 \tag{11}$$

The temperature dependence of the stress is investigated using Equation 12. The resulting derivatives of the disengagement time with respect to the temperature (D1) and of the stress with respect to the disengagement time (D2) are investigated separately for a possible change of their sign.

$$\frac{d\sigma}{dT} \approx \frac{\Delta\sigma}{\Delta T} = \frac{\Delta\tau_d\,\Delta\sigma}{\Delta T\,\Delta\tau_d} \approx \underbrace{\frac{d\tau_d}{dT}}_{D1}\underbrace{\frac{d\sigma}{d\tau_d}}_{D2} < 0 \tag{12}$$

The aim of this consideration is to identify the conditions under which Equation 11 is fulfilled, so that the constitutive material model equation of reptation theory (Equation 4) predicts a decreasing stress with increasing temperatures, as observed in the experiments.

Investigation of D1: The derivative of the disengagement time with respect to the temperature is given in Equation 13 using Equation 1.

$$\frac{d\tau_d}{dT} = \frac{d}{dT}\left(\frac{20\,M_e}{\pi^2 RT\rho}\eta_0(T)\right) \tag{13}$$

As shown in Figure 8, the zero viscosity decreases monotonously with increasing temperature. The material properties M_e and ρ are considered temperature independent. The temperature in the denominator has a negligible influence on the result compared to the zero viscosity. For this reason, it is expected that the derivative in Equation 13 will always be negative. Table 4 shows the evaluation of the derivate of the disengagement time with respect to the temperature according to Equation 13 at different temperatures.

Temperature [°C]	$d\tau_d/dT$ [s/K]
80	-1118.03
90	-4.61
100	-6.69E-02
110	-2.30E-03
120	-1.47E-04
130	-1.47E-05
140	-2.10E-06

Table 4: The values of the derivative of the disengagement time with respect to the temperature

Table 4 shows that in the considered temperature range between 80 and 120 °C, the values decrease strictly monotonously with increasing temperatures and thus, the sign of the derivate remains negative:

$$\left.\frac{d\tau_d}{dT}\right|_{T\epsilon[80,140]} < 0 \tag{15}$$

Investigation of D2: The derivative of the stress with

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respect to the disengagement time is given in Equation 16 using the Equation 4.

$$\frac{d\sigma_{\alpha\beta}^{(P)}(t)}{d\tau_d} = \frac{d}{d\tau_d}\left[G_e \int_{-\infty}^{t} \frac{d}{dt'}\psi(t-t')\mathcal{Q}_{\alpha\beta}^{(IA)}(E(t,t'))dt'\right] \quad (16)$$

The plateau modulus G_e is assumed as a constant, material specific value and the orientation tensor $\mathcal{Q}_{\alpha\beta}^{(IA)}$ depends solely on the deformation history, but not on the material properties of the polymer. Thus, these quantities do not depend on the disengagement time. The Doi-Edwards memory function ψ and $\mathcal{Q}_{\alpha\beta}^{(IA)}$ are continuously differentiable functions and thus, the derivative of the integrand is equal to the derivative of the integral. This leads to Equation 17, in which the derivative of the Doi-Edwards memory function ψ is the only quantity depending on the disengagement time.

$$\frac{d\sigma_{\alpha\beta}^{(P)}(t)}{d\tau_d} = G_e \int_{-\infty}^{t} \frac{d}{\tau_d}\left[\frac{d}{dt'}\psi(t-t')\right]\mathcal{Q}_{\alpha\beta}^{(IA)}(E(t,t'))dt' \quad (17)$$

The derivative of the Doi-Edwards memory function is given in Equation 18.

$$\frac{d}{dt'}\psi(t-t') = \sum_{p:\,\mathrm{odd}} \frac{8}{\pi^2\tau_d} exp\left(-\frac{(t-t')\,p^2}{\tau_d}\right) \quad (18)$$

The derivative of Equation 18 with respect to the disengagement time results in Equation 19.

$$\frac{d}{d\tau_d}\left(\sum_{p:\,\mathrm{odd}} \frac{8}{\pi^2\tau_d} exp\left(-\frac{(t-t')p^2}{\tau_d}\right)\right) = \sum_{p:\,\mathrm{odd}} \frac{8}{\pi^2} \frac{exp\left(-\frac{(t-t')p^2}{\tau_d}\right)}{\tau_d^3}\left((t-t')p^2 - \tau_d\right) \quad (19)$$

Equation 19 is not strictly positive for all values of $\tau_d > 0$ and $(t-t') > 0$ but depends on the ratio of τ_d and $(t-t')$, which is expressed by n':

$$n' = \frac{(t-t')}{\tau_d} \Rightarrow n'\tau_d = (t-t') \quad (20)$$

Using the ratio n' from Equation 20, Equation 19 is examined for a zero point for different values of n':

$$\sum_{p:\,\mathrm{odd}} exp(-n'p^2)(n'p^2 - 1) = 0 \quad (21)$$

Figure 14 shows the evaluation of Equation 21 for different values of n'. The zero point is located at $n' \approx 0.99727 = n_0$.

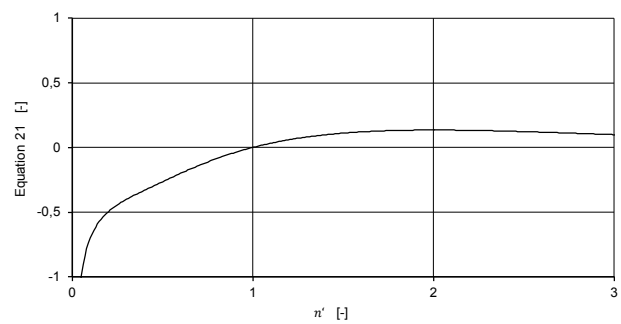

Fig. 14: Evaluation of Equation 21 with a zero point at $n' \approx 0.99727$

The evaluation of Equation 21 shows that the sign of the derivate of the Doi-Edwards memory function in Equation 19 remains negative for $n' < n_0$:

$$\frac{d}{d\tau_d}\left(\frac{d}{dt'}\psi(t-t')\right)\Bigg|_{n' < n_0} < 0 \Rightarrow \frac{d}{d\tau_d}\left(\frac{d}{dt'}\psi(t-t')\right)\Bigg|_{\tau_d \in \left[\frac{(t-t')}{n_0},\infty\right]} < 0 \quad (22)$$

For the considered time range of $t' \in [0,t]$, $\tau_d \geq t/n_0$ fulfils Equation 22 and subsequently Equation 23 holds under this condition.

$$\frac{d\sigma_{\alpha\beta}^{(P)}(t)}{d\tau_d} < 0 \quad \text{if} \quad \tau_d \geq \frac{t}{n_0} \quad (23)$$

Investigation of Equation 12: The investigations of terms D1 and D2 have shown that D1 is strictly negative in the considered temperature range $T \epsilon [80,140]$. In contrast, the term D2 can be positive or negative depending on the ratio of the disengagement time to the experiment time. The investigation of D2 has shown, that the term has a negative value for disengagement times exceeding the value t/n_0 and thus Equation 24 holds. If the disengagement time is less than t/n_0, D2 may remain negative or become positive for $\tau_d \ll t/n_0$.

$$\underbrace{\frac{d\tau_d}{dT}}_{<0}\underbrace{\frac{d\sigma}{d\tau_d}}_{<0} > 0 \quad \text{if} \quad \tau_d \geq \frac{t}{n_0} \quad (24)$$

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High disengagement times resulting from processing temperatures slightly above the glass temperature range and short stretching times due to high strain rates, as it is the case in the two-stage stretch blow moulding process, exceed the validity limits of reptation theory. In case of the considered biaxial stress-strain curves at 100 °C at a strain rate of 1/s, the disengagement time τ_d is 49.58 s and the experiment time t is 2.6 s. Therefore $\tau_d \gg t/n_0$ and thus, the model cannot be applied at the processing conditions of the two-stage stretch blow moulding process.

Conclusion and Outlook

The aim of this paper is to investigate the applicability of reptation model for predicting the stress-strain behaviour of biaxial stretched PET slightly above the glass transition temperatures at high strain rates, using the experimental strain-stress curves as a basis for comparison.

The material properties required for the stress equation are determined using various measurement methods and literature values. The measurement of the plateau modulus and the zero shear viscosity cannot be achieved with the same accuracy as in the case of polymer melts, since the measurement methods used have shown instabilities in the glass transition temperature range, which in turn led to high measurement uncertainties. Thus, the measurement of the parameters and the application of the model has shown that new measurement approaches for the reproducible determination of the parameters and new approaches for the components of the model are required for the accurate prediction of the stress of highly viscous oriented polymers caused by biaxial strain.

The application of the model has shown deficiencies in the quantitative agreement to biaxial stretch tests, however, qualitatively, the model reproduces the stress-strain behaviour for strains up to 100%. The modelled viscoelasticity represented through the orientation tensor was in the correct order of magnitude, but was underestimated. In case of the temperature dependency, the model results are neither qualitatively nor quantitatively correct. The investigations indicate that the temperature dependency is predicted incorrectly. This behaviour has been further investigated. As the experiments show, an increasing temperature leads to a decreasing stress and thus the investigations consider the derivative of the stress with respect to the temperature. It is shown that the derivative of the Doi-Edwards memory function has an inflexion point if the stretching duration equals the disengagement time ($t \approx \tau_d$). For very small disengagement times compared to the stretching duration ($\tau_d \ll t$), the results of the model would match the experimental tests, but for high disengagement times induced through the large viscosity near the glass transition temperature and low stretching times induced through high strain rates ($\tau_d \gg t$), the Doi-Edwards model equation cannot be applied.

Concluding, the measurement of the parameters and the application of the model has shown that new measurement approaches for the reproducible determination of the parameters and new approaches for the components of the model are required for the accurate prediction of the stress of highly viscous oriented polymers caused by biaxial strain slightly above the glass transition temperature. For the investigated model, the investigations have shown that the process conditions of the two-stage stretch blow moulding process exceed the validity limits of reptation theory.

In this paper a criterion could be defined which shows the limitation of the reptation theory for forming processes where high strain rates as well as high viscosities are present. This criterion, however, does not exclude all forming processes, so that the material properties of the melt in the extrusion blow moulding process may be predicted with reptation model.

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